

The Mental Health Impact of Youth Unemployment in the Netherlands

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2025-05-26

Set-up your environment

```
require(tidyverse)
```

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Tutorial 2

Chantal Schouwenaar

Introduction

1.1 Describe the Social Problem

In the Netherlands, depression has proven to be the most prevalent mental health disorder, with almost one in five individuals expressing depressive symptoms. Unemployment has been identified as a main factor that aggravates this issue. Beyond its financial implications, being out of work has substantial psychological consequences. It often has an effect on self-esteem, increased physical and cognitive issues, and, in severe cases, suicidal tendencies (Ringdal & Rootjes, 2022). These aspects are not only a result of economic stress but also a loss of other benefits such as a sense of purpose, the absence of routine, and decreased social connections (ElMorsy, 2024). There is an extensive amount of research surrounding the correlation between mental health and unemployment. This analysis is focused on how this relationship affects young people in the Netherlands. For emerging adults, working is important as it is associated with obtaining greater independence, forming an identity, and starting a career. In the Netherlands, individuals are legally allowed to work from the age of 13 with certain conditions and freely from the age of 16 (Netherlands Enterprise Agency, n.d). Employment from a young age is widely normalized and many times expected during adolescence. Therefore, not having a job can be unsettling and disruption of deeply rooted social expectations, leading to a higher risk of mental health implications.

Part 2 - Data Sourcing

2.1 Load in the data

```
#import of mental data gemeentes 2020
Aatillgoereemental2020 = read.csv("data/Aatillgoereemental2020.csv")
goestillnissewaardmental2020 = read.csv("data/goestillnissewaardmental2020.csv")
noardtillvlielandmental2020 = read.csv("data/noardtillvlielandmental2020.csv")
vlissingentillzwollemental2020 = read.csv("data/vlissingentillzwollemental2020.csv")

#import of mental data gemeentes 2022
Aatillgoereemental2022 = read.csv("data/Aatillgoereemental.csv")
goestillnissewaardmental2022 = read.csv("data/goestillnissewaardmental2022.csv")
noardtillvlielandmental2022 = read.csv("data/noardtillvlielandmental2022.csv")
vlissingentillzwollemental2022 = read.csv("data/vlissingentillzwollemental2022.csv")

#import of unemployment data 2020-2023
gem_unemployment = read.csv("standard data/gem_unemployment.csv")
```

2.2.1 Provide a short summary of the dataset(s)

For this project, we used all the gemeentes in the Netherlands, that could be roughly considered municipalities. Our data came from the national government's website (CBS StatLine). There is missing data because there was no information available for certain years - item 2.3 explains in more detail each variable of our analysis.

Regarding the unemployment data: the observed population was registered in the Basisregistratie Personen (BRP) as of October 1st of the reference year, which is the date when data is collected yearly. The amount of people going to school, for example, is determined based on their status on October 1st. The classification of the municipalities is based on their situation as of January 1st, 2024. The data on individuals looking for a job and registered with the UWV comes from two sources: from 2005 to 2018, it is taken from UWV Werkbedrijf, and from 2019 onwards, it is based on the Geregistreerd Werkzoekende UWV system. All the numbers are absolute. Regarding the mental health data: the information is retrieved from a model of calculation used by the National Institute for Public Health and the Environment (RIVM) based on data from the Health Monitor for Adults and the Elderly for the year 2020 and the the Corona Health Monitor 2022 from GGDs, CBS and RIVM. Every four years, new data from the Health Monitor for Adults and the Elderly become available and that is obtained through surveys with an anonymous nature. The format of the questions follows the MHI-5 and the Kessler-10 (K10) questionnaires.

We choose to upload the head of the dataset from RStudio only after performing data cleaning, as the original datasets are highly unstructured and not easily readable in their raw state.

2.3 Describe the type of variables included

MHI-5: Percentage of people reporting psychological complaints

The analysis surrounding psychological complaints is measured by the Mental Health Inventory 5 (MHI-5), which is the international standard procedure to measure mental health. It includes five questions related to the well-being of an individual over the past month, related to displays of anxiety, depression, behavioral/emotional control, and general positive effects. Based on the results of the questionnaire, people received a score ranging from 0 (extremely unhealthy) to 100 (perfectly healthy). A person is considered psychologically unhealthy by the Statistics Netherlands (CBS) if they have received a score lower than 60.

The only data we had available was for the year 2022 because the data for 2020, 2021, and 2023 were not available and that explains the N/A markings in the graph.

High Risk Anxiety and Depression: Percentage of people aged 18 or older at high risk of anxiety disorder or depression

The data was based on the Kessler-10 (K10) questionnaire: a very commonly used screening questionnaire for anxiety and depression. It includes 10 questions concerning emotional states like worthlessness, tiredness, and nervousness (during the past 4 weeks). Each question is answered on a 5-point scale, ranging from “Always” (highest score of 5) to “Never” (lowest score of 1). If 3 or more answers are marked as “Impossible to answer”, the indicator receives the value Impossible. If 1 or 2 items are Impossible, their values are estimated based on the average score for those items. The total score ranges from 10 to 50, where 10-50 suggests that there’s no or low risk, 16-29 that there is moderate risk, and 30-50 that there is a high risk of anxiety disorder or depression. We could find data for 2020 and 2022.

Unemployment Variables

The data inside the table shows the labor market situation of young people who are between 15 and 27 years old in the Netherlands (total youth), who were registered in the Basisregistratie Personen (BRP). It indicates whether a person is working, studying, searching for work, and registered as a job seeker at the UWV, or receiving benefits. The total unemployed youth represents the people who are not actively looking for a job or cannot find work. The people working can be divided into self-employed or employees. This data can be split into different regions.

Part 3 - Quantifying

3.1 Data cleaning

3.1.1 Make 1 large mental data set of all the small partial mental datasets for 2020 and 2022

3.1.2 Get rid of the useless variables in mental data set

3.1.3 Get rid of the useless variables in unemployment data set

3.1.4 Prepare mental and unemployment set so that we can merge them together in 1 data set

3.1.5 Change unemployed from a row variable to a column variable

3.1.6 Now merge the 2 data sets together

3.1.7 Clean the combined data set so that all useless variables are filtered out

2.2.2 Provide a short summary of the dataset(s)

```
head(Data_Cleancombined)
```

##	Periode	Gemeente	Aantal_jongeren	Niet_werkzame_jongeren	Regio_type
## 1	2020	Nederland	2586000	818900	Land
## 2	2020	Aa en Hunze	2900	900	Gemeente

```
## 3      2020      Aalsmeer      4500      1200 Gemeente
## 4      2020      Aalten      3600      800 Gemeente
## 5      2020 Achtkarspelen      4000      1200 Gemeente
## 6      2020 Alblasserdam      2900      700 Gemeente
##      Psychische_klachten Hoog_risico_angst_depressie
## 1              NA              6.8
## 2              NA              3.7
## 3              NA              4.9
## 4              NA              5.4
## 5              NA              4.6
## 6              NA              6.5
```

```
str(Data_Cleancombined)
```

```
## 'data.frame':  1398 obs. of  7 variables:
## $ Periode      : chr  "2020" "2020" "2020" "2020" ...
## $ Gemeente     : chr  "Nederland" "Aa en Hunze" "Aalsmeer" "Aalten" ...
## $ Aantal_jongeren : int  2586000 2900 4500 3600 4000 2900 3300 14800 10800 32400 ...
## $ Niet_werkzame_jongeren : int  818900 900 1200 800 1200 700 1100 4000 3500 11500 ...
## $ Regio_type    : chr  "Land" " " "Gemeente" " " "Gemeente" " " "Gemeente" " " ...
## $ Psychische_klachten : num  NA NA NA NA NA NA NA NA NA NA NA ...
## $ Hoog_risico_angst_depressie: num  6.8 3.7 4.9 5.4 4.6 6.5 5.5 6.9 7.7 7.6 ...
```

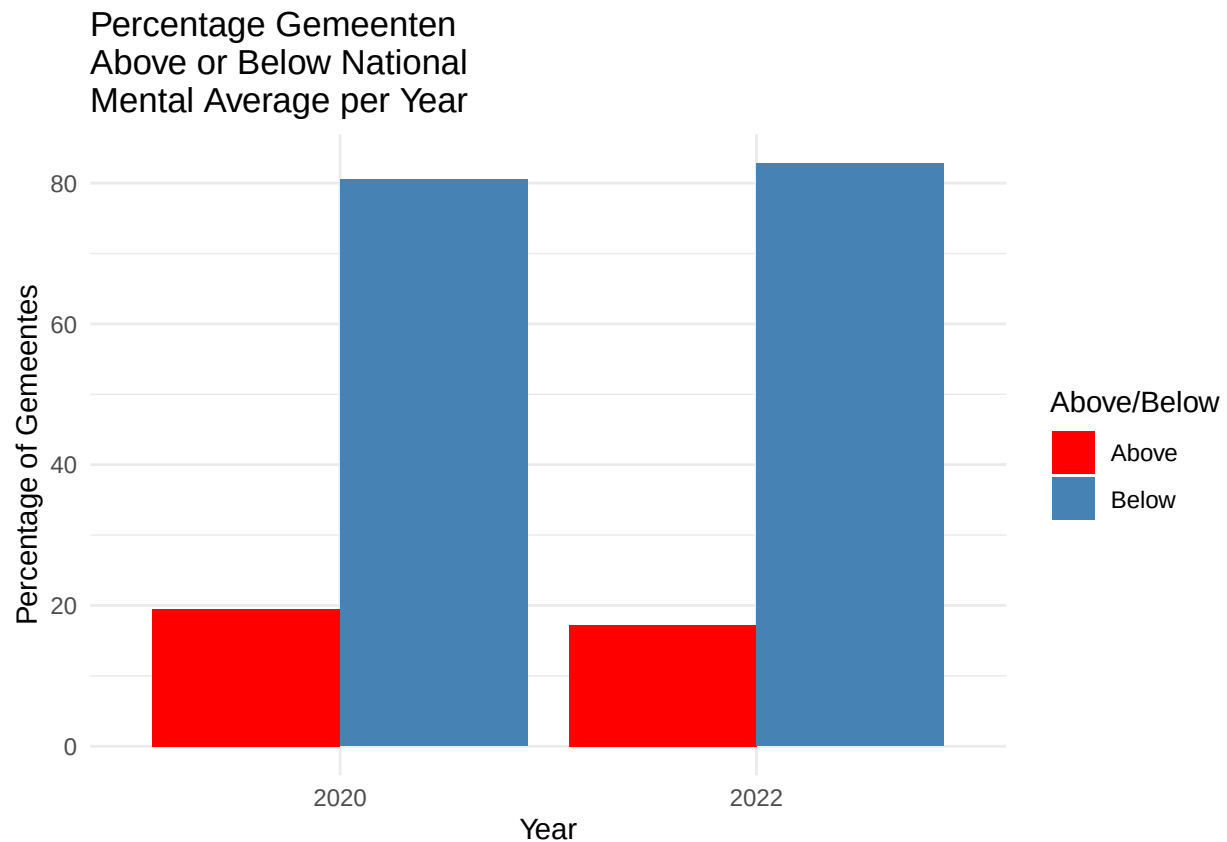
So in summary, the cleaned dataset includes the absolute numbers of both young people and unemployed individuals. Unemployment figures are presented for the period from 2020 to 2023, while the risk of anxiety and depression is shown for the years 2020 and 2022.

3.2 Generate necessary variables

Variable 1: Unemployment percentage

```
Data_Cleancombined$unemployment_percentage = (Data_Cleancombined$Niet_werkzame_jongeren/Data_Cleancombined$Aantal_jongeren)*100
Data_Cleancombined$unemployment_percentage = round(Data_Cleancombined$unemployment_percentage,2)
```

Variable 2: Above or Below the national average risk of anxiety and depression



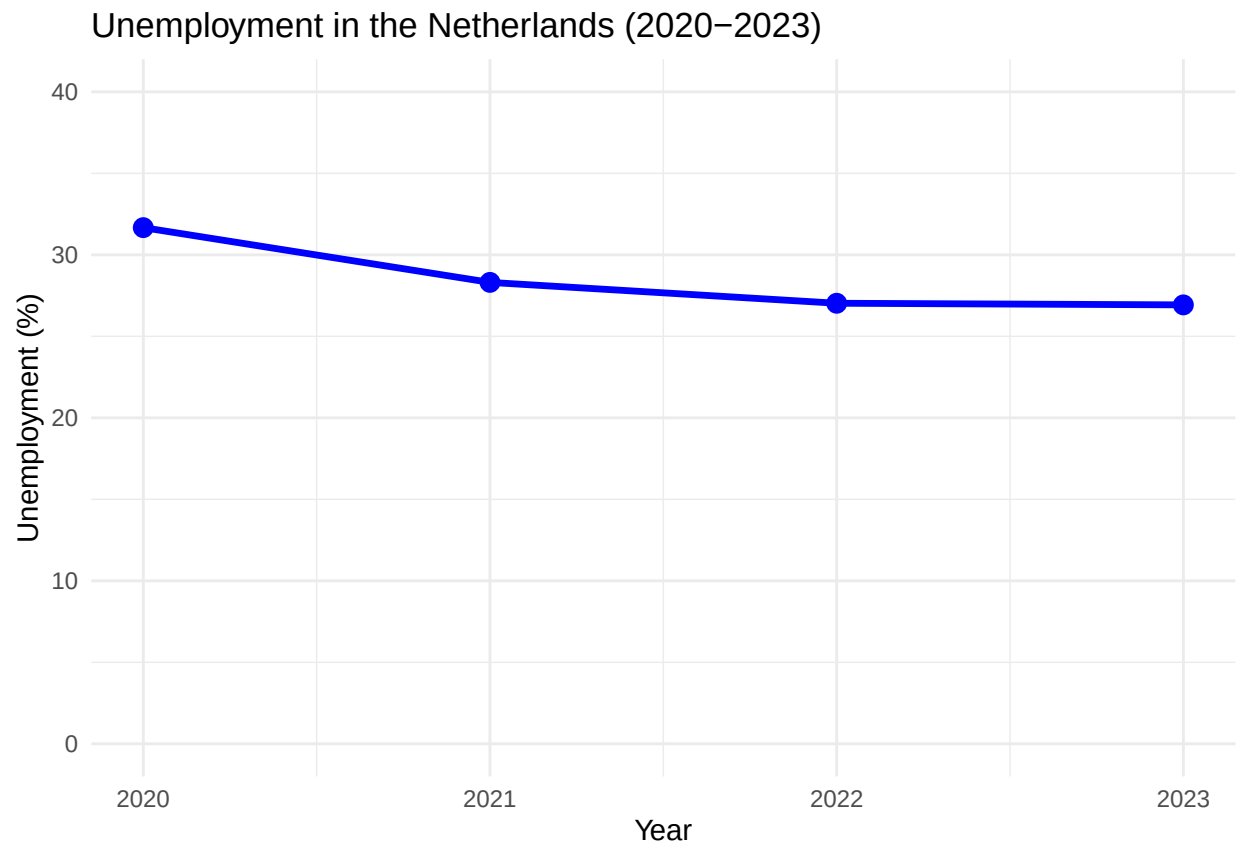
Variable 3: Above or Below the national average unemployment

Variable 4: Unemployment Percentage Change

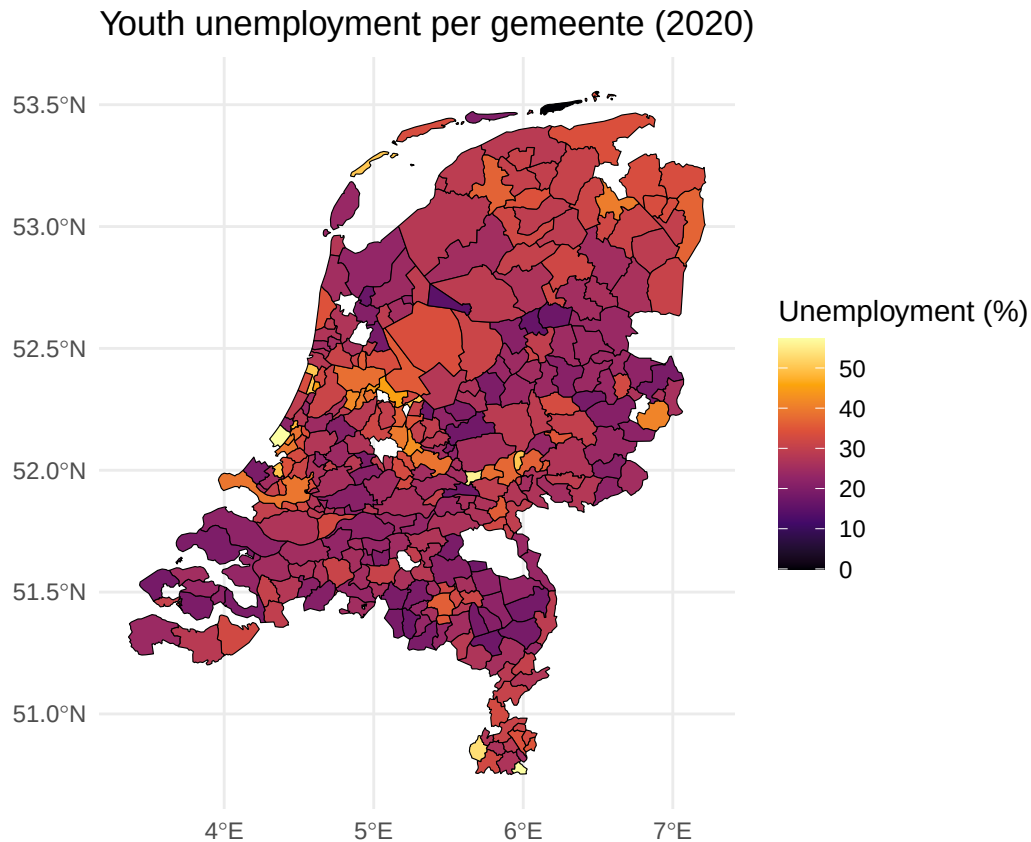
```
###unemployment percentage change
Data_Cleancombined$Periode = as.numeric(Data_Cleancombined$Periode)

Data_Cleancombined <- Data_Cleancombined %>%
  arrange(Gemeente, Periode) %>%
  group_by(Gemeente) %>%
  mutate(
    unemployment_change = if_else(
      Periode - lag(Periode) == 1 & !is.na(lag(Niet_werkzame_jongeren)),
      (Niet_werkzame_jongeren - lag(Niet_werkzame_jongeren)) / lag(Niet_werkzame_jongeren) * 100,
      NA_real_
    )
  ) %>%
  ungroup()
```

3.3 Visualize temporal variation



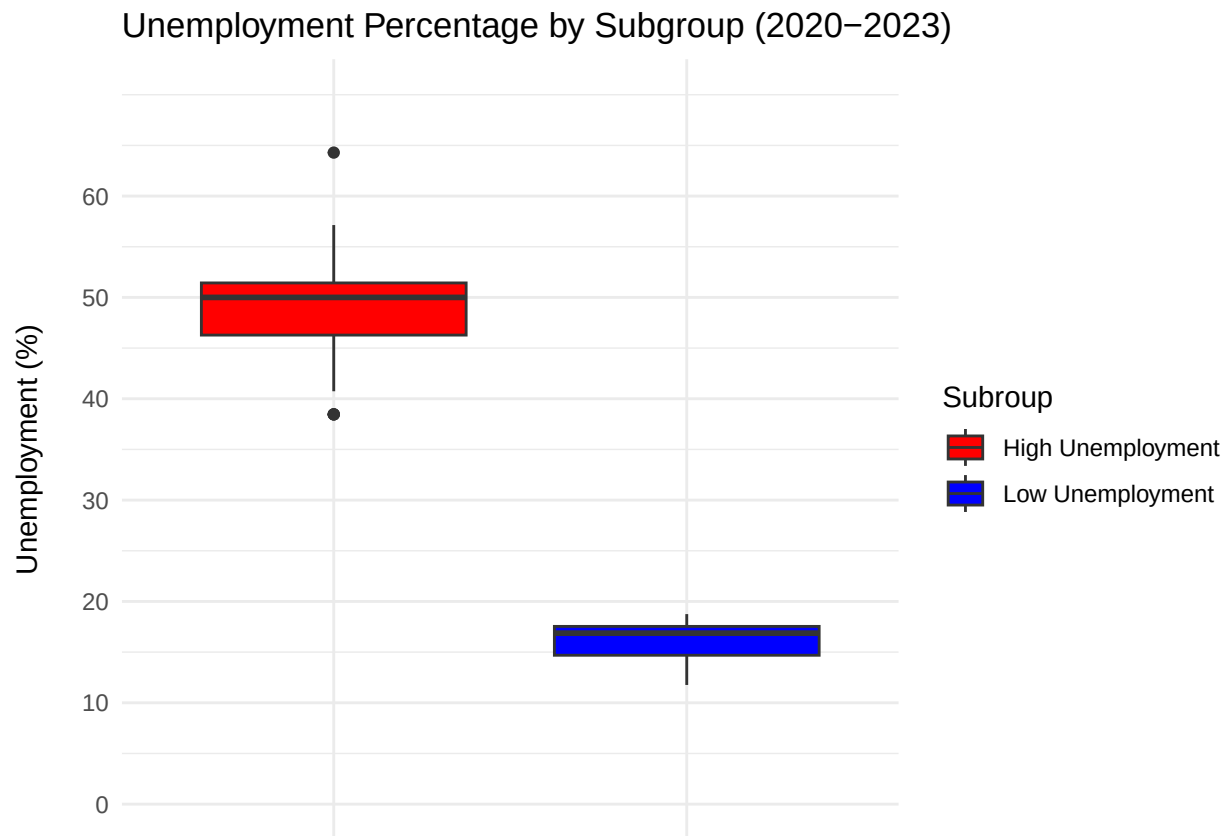
3.4 Visualize spatial variation



The heat map above shows the percentage of youth unemployment per Dutch municipality (gemeente) in the year 2020, ranging from 0% to over 50%. This spatial variation is important for our analysis as it allows us to better identify regional disparities in youth unemployment across the Netherlands. By making more apparent the municipalities with particularly high or low unemployment rates, where there may be a greater risk of mental health issues among young people. It helps us with the overall understanding of the multitude of factors that may have an impact on the social problem overall. The contrasting data can be explained by the very low population in some areas, which then has a stronger impact on the general percentage.

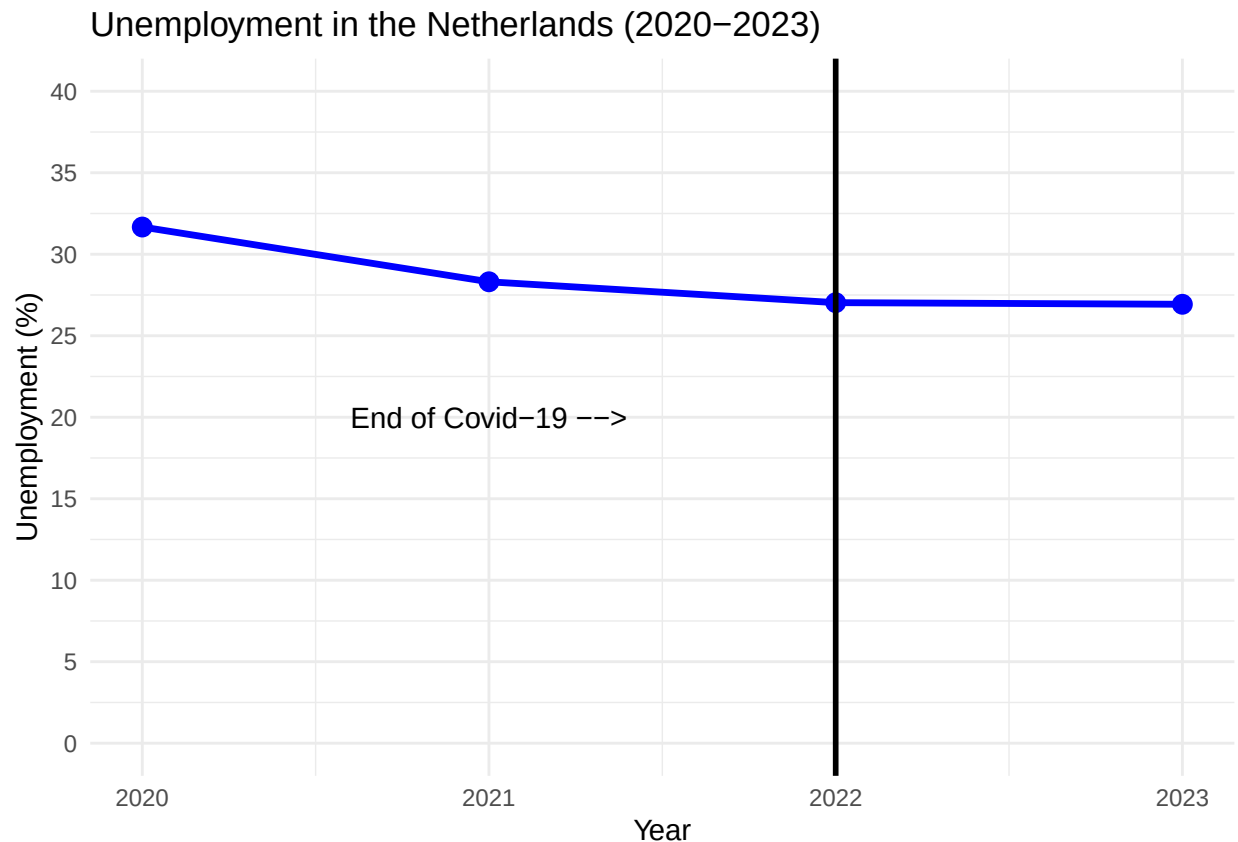
Because the municipalities in the geographical dataset did not fully match those in our own dataset, some areas appear blank (in white) on the map. This discrepancy is mainly due to differences in naming conventions and the fact that municipalities are frequently merged or split over time.

3.5 Visualize sub-population variation



The box plot above compares unemployment percentages from 2020 to 2023 between two subgroups: the 10 municipalities with the highest youth unemployment and the 10 with the lowest. The municipalities Vlieland and Schiermonnikoog were excluded due to the fact that they are islands and have extremely small populations, which affected the accuracy of the overall results by showing zero unemployment in some years. This visualization is essential to our analysis of youth unemployment and its relationship to mental health because it illustrates a clear divide between the two subgroups, with the high-unemployment group consistently showing rates near or above 50%, while the low-unemployment group remains closer to 15%. This disparity helps us analyze the concept of economic pressure and its implications on psychological well-being.

3.6 Event analysis



The graph above displays the youth unemployment in the Netherlands from the years 2020 to 2023, highlighting the end of the COVID-19 pandemic in 2022. This event is crucial when analyzing our main topic because the pandemic affected both the labor market and young people's mental well-being. During 2020 and 2021, there were higher rates of unemployment, since a lot of businesses had to adapt financially to the new reality. Entry-level jobs and part-time jobs, occupied by many young people, suffered cuts. After the COVID-19 lockdowns, it is possible to see a recovery, but the unemployment rates remained high since.

Part 4 - Discussion

Between 2020 and 2023, unemployment levels in the Netherlands gradually declined, but at varying rates. In 2020, unemployment peaked at around 34%, likely due to the economic instability caused by the Covid-19 pandemic. By 2022, as the pandemic's effects started to calm and economies began to recover, unemployment levels started to stabilize.

Our analysis revealed a significant disparity in unemployment rates. The High Unemployment subgroup has a median of around 50% and the Low Unemployment subgroup has a median of around 16%. Gemeentes with high unemployment have more than triple the rates of the low unemployment subgroup, which tells us that there is a sharp divide between the two subgroups.

We also observed noticeable geographic patterns in youth unemployment. Higher unemployment (30% to 50% and higher) is concentrated in certain gemeentes, particularly in Heerlen, Rotterdam, Tilburg, and others, which we can see is clearly yellow. Lower unemployment (0 to 10% and under) appears to be more frequent in smaller municipalities in Gelderland and Overijssel, some parts of North Brabant and Zeeland and Northern Friesland. Another interesting statistic is the high unemployment rate in the municipalities

of Delft and Wageningen. This can be logically explained by the fact that these are well-known student cities. Many students do not have time to work alongside their studies, which contributes to the higher youth unemployment rates in these areas.

Regarding mental health, our results show that most gemeentes reported better than the national average mental health. Each year, from 2020 to 2023, around 76 to 80% of gemeentes were below the national average, and around 20 to 25% were above. That indicates that mental health issues are more concentrated in only a small portion of the total gemeentes. The percentage of gemeentes above the national average is decreasing while the percentage below is increasing. While it suggests that fewer areas are facing difficulties, the ones that are may be experiencing more severe challenges.

Overall, our findings suggest a negative correlation between unemployment and mental health. As unemployment declined from 2020 to 2023, mental health levels across the country improved. This supports the understanding that economic stability and employment are linked to one's mental well-being.

Part 5 - Reproducibility

5.1 Github repository link

<https://github.com/Collinveenstra/Programming-group2>- Both the raw and cleaned datasets, along with the script, can be found in the folder: "final project"

5.2 Reference list

Centraal Bureau voor de Statistiek. (2024). CBS Statline. Rijksinstituut voor Volksgezondheid en Milieu. Retrieved from: <https://statline.rivm.nl/#/RIVM/nl/dataset/50120NED/table>

Centraal Bureau voor de Statistiek. (2025). CBS Statline. Retrieved from: <https://opendata.cbs.nl/statline/#/CBS/nl/dataset/86087NED/table>

ElMorsy, M. (2024, January 23). The silent struggle: The impact of unemployment on mental health. Career Edge. Retrieved from: <https://www.careeredge.ca/the-silent-struggle-the-impact-of-unemployment-on-mental-health/>

Netherlands Enterprise Agency. (n.d.). Employment of minors and young adults. Business.gov.nl. Retrieved from: <https://business.gov.nl/regulation/employment-young-people/>

Ringdal, C., & Rootjes, F. (2022). Depression and labor supply: Evidence from the Netherlands. *Economics & Human Biology*, 45, 101103. Retrieved from: <https://doi.org/10.1016/j.ehb.2021.101103>